



NTNU – Trondheim
Norwegian University of
Science and Technology

TET4185: Power Markets, Resources and Environment

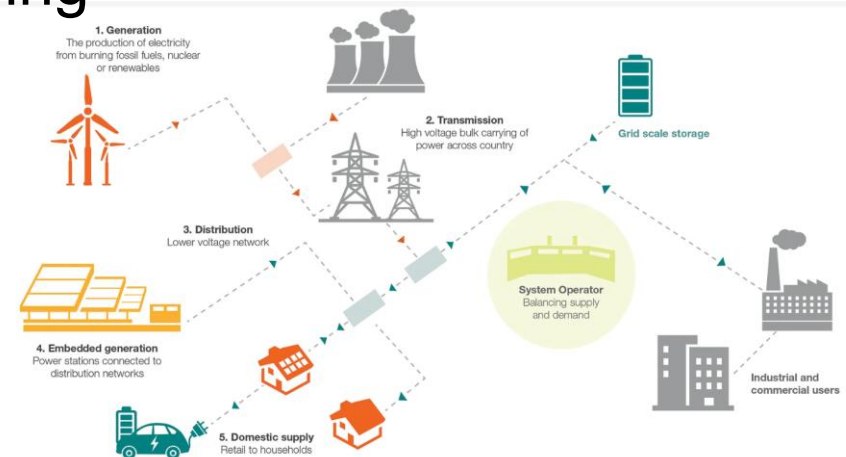
Summing up

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Professor
Spring 2026



The overview of the course:

- Basic of microeconomics
- Market for Electrical Energy Participating in the Market
- Pricing & Restructuring and Nordic Market
- Grid access, Congestion management
- Transmission pricing based on OPF
- Grid-constrained market clearing
- Flow-Based Market Coupling
- Environmental policy
- Hydro Generation
- Ancillary services



Source: <https://www.energy-uk.org.uk/insights/different-parts-of-the-energy-market/>

The topics of this course are becoming more and more relevant



Strømstøtte som
fordeling av grunnrente

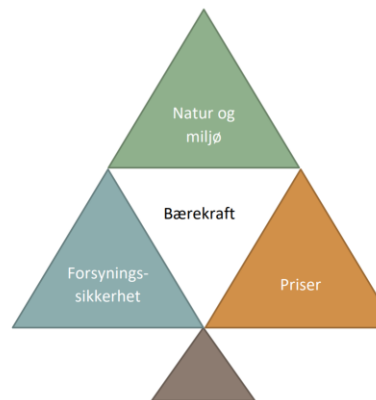
Et forslag til en mer klimavennlig og
rettferdig strømstøtteordning

Strømprisene stiger – Sør-Norge får
årets høyeste pris i dag



Det er ventet høye strømpriser i Sør-Norge onsdag. FOTO: FOTO: LISE ÅSERUD / NTB

Balansekunst



Rapport fra Strømprisutvalget, 12. oktober 2023



Norsk klimastiftelse
NORGES GRØNNE TANKESMIE

Klimaråd 2025
Vurderinger og anbefalinger for en
bedre klimapolitikk i Norge

Det er behov for et taksakfte i norsk klimapolitikk. Derfor etablerte Norsk klimastiftelse høsten 2024 en ekspertgruppe, Klimaråd 2025. Medlemmene er blant landets fremste eksperter på klima-, natur- og energisaker. De fikk i oppgave å vurdere Norges klimastilling og komme med anbefalinger til veien videre. Denne rapporten oppsummerer deres vurderinger og anbefalinger.

NR. 3 / 2025

ENERGIWATCH

FORNYBAR NETT & TEKNOLOGI OFFSHORE OLJE & GASS POLITIKK & MARKED KARRIERE

POLITIKK & MARKED | ÅRSOVERSIKT

Flytbasert markedskobling skaper
«kontraintuitiv flyt»: - Vi ser
hvordan spotprisene har gått
bananas

Prismåte NCS importerte over dobbelt så mye som de eksporterte, men fikk likevel Nordens soleklart laveste strømpriser.



Flytbasert markedskobling har ført til anslupper som er krevende å forstå for markedsdeltakere. | Foto: Nord Pool

Basic microeconomics

(sources: IW (Ch. 2), ecture Slides, Assignment 1)

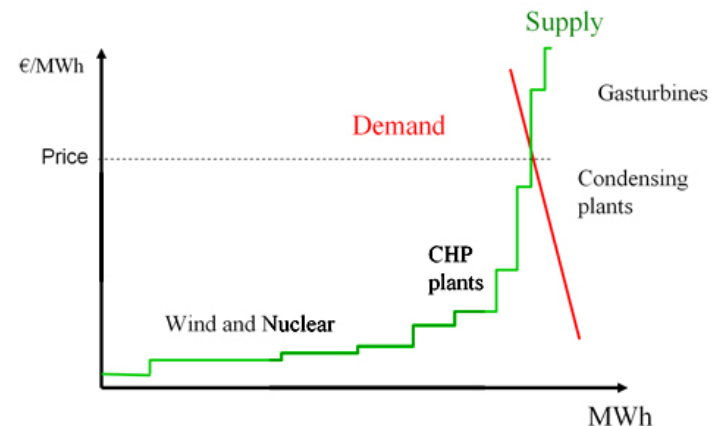
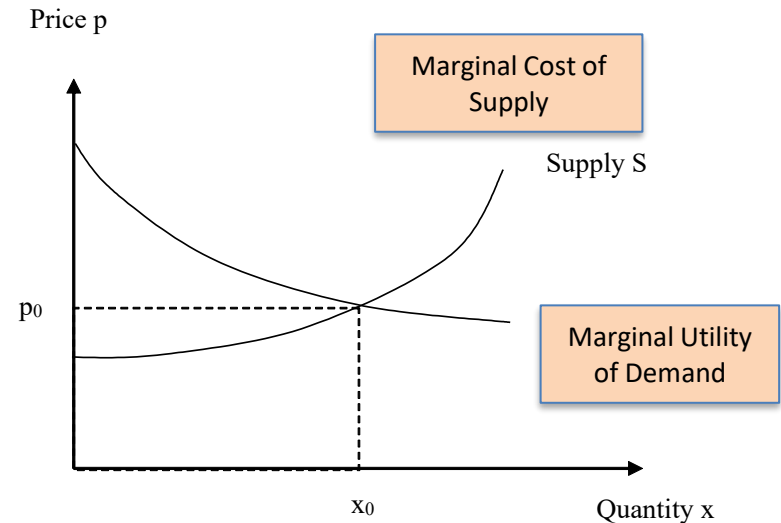
- Supply-demand cross, social welfare, consumer and producer surplus
- Definition and use of demand elasticity
- The relation between total cost, average cost and marginal cost
- Different form of competition:
 - perfect
 - monopolistic
 - oligopolistic (Cournot)
- Be able to calculate quantities and prices in simple examples of non-perfect competition

IW: Ivar Wangensteen, “Power System Economics: The Nordic Electricity Market”, Second Edition, 2012.

DK: Daniel S. Kirschen and Goran Strbac, “Fundamentals of Power System Economics”, John Wiley & Sons, 2004.

Marshallian supply-demand cross

- Alfred Marshall:
 - «Supply and demand interact to determine market balance, deciding price and quantity»
- The market cross is «equilibrium»
 - One defined price for a resource
- Practically, the trading is done to lead to one unified price



Market for Electrical & Energy Participating in the Market

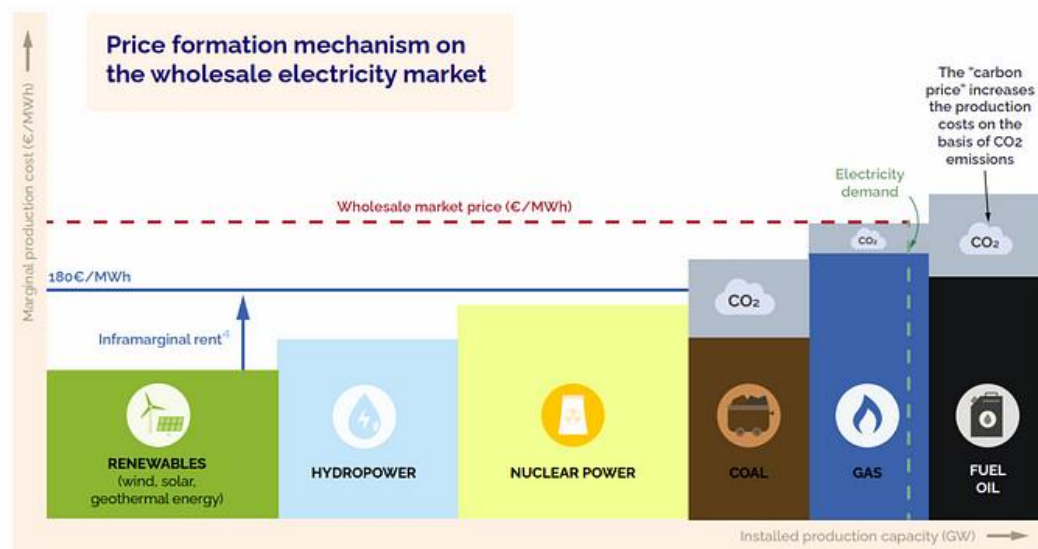
(sources: DK (Ch. 3&4), Lecture Slides, Assignment 2)

- Different types of electricity trading (pool and bilateral trading)
- Different types of electricity market (financial, spot, intraday, balancing)
- Participating in electricity markets: the producer's perspective
 - Marginal, infra-marginal, extra-marginal producers
 - Bidding under perfect competition
 - Optimal Scheduling (start up cost, temporal constraints and environmental constraints)
- Participating in electricity markets: the consumer's perspective and retailer's perspective

DK: Daniel S. Kirschen and Goran Strbac, "Fundamentals of Power System Economics", John Wiley & Sons, 2004.

Electricity Market

- A market is an environment to help buyers and sellers interact and agree on transactions
- Market equilibrium: the price clears the market, that is, the supply is equal to demand
- Large consumers and retailers purchase electrical energy from generating companies based on their forecasted consumption



Pricing

(sources: IW (Ch. 4), Lecture Slides, Assignment 3)

Key Concepts:

- Explore tariffs and their economic implications.
- Understand the effects of market restructuring and unbundling.
- Analyze different contract types and grid tariff structures.
- Investigate Ramsey Pricing and its application in setting grid tariffs.

Goals:

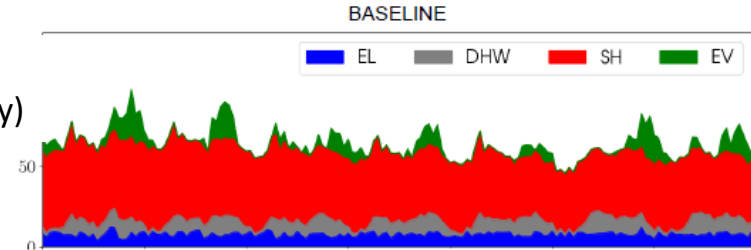
- Identify the challenges and benefits of various pricing mechanisms.
- Evaluate the impact on consumers and the energy market, with a focus on the efficiency and equity of Ramsey Pricing.

IW: Ivar Wangensteen, “Power System Economics: The Nordic Electricity Market”, Second Edition, 2012.

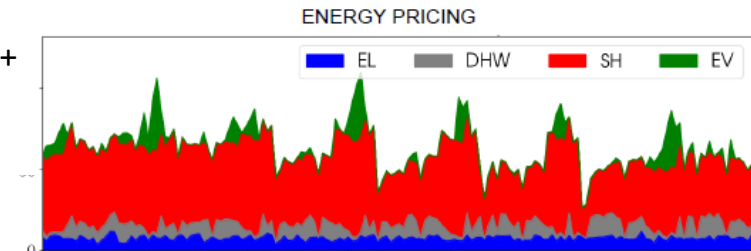
Flexibility in an apartment block

- Price-incentive decides the potential for
 - Load shifting
 - Peak load reduction
- Price-incentive consists of
 - Combination of spot price and grid tariff
 - New: flexibility markets?

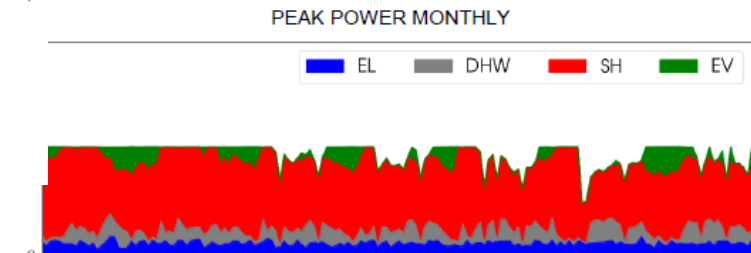
Baseline
(without flexibility)



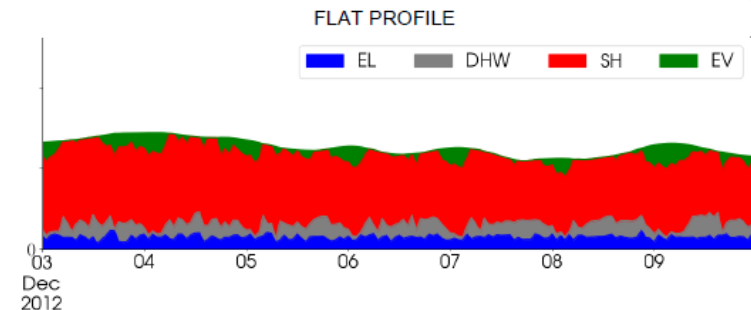
Energy grid tariff +
spot price



Monthly peak
tariff
+ spot price



Flat load profile
(regardless of
price)



Kilde: SINTEFs beregninger i FlexBuild-p
<https://www.sintef.no/projectweb/flexbuild/>

Restructuring and Nordic Market

(sources: IW (Ch. 5), sources: IW (Ch. 8)- Optimal mix of generation units, Lecture Slides, Assignment 4)

- Analyze the transition from regulated to deregulated markets and its impact on pricing, efficiency, and security of supply.
- Understand the roles and responsibilities in deregulated electricity markets, including power exchanges, regulators, and market participants.
- Screening model: understand and be able to apply the simple screening model for the optimal generation mix
 - The missing money problem
- Market clearing as an optimization Problem

IW: Ivar Wangensteen, “Power System Economics: The Nordic Electricity Market”, Second Edition, 2012.

Centralized social welfare optimization

- The social welfare maximization problem can be written as

$$\begin{aligned} \text{Max}_{p_g^G \geq 0, p_d^D \geq 0} \quad SW = & \underbrace{\sum_{d=1}^{N_D} U_d p_d^D}_{\text{Utility of demands}} - \underbrace{\sum_{g=1}^{N_G} C_g p_g^G}_{\text{Cost of electricity production}} \quad (1a) \end{aligned}$$

subject to:

$$\sum_{d=1}^{N_D} p_d^D - \sum_{g=1}^{N_G} p_g^G = 0 \quad (\lambda) \quad (1b) \rightarrow \text{Power balance}$$

$$0 \leq p_g^G \leq \bar{P}_g^G \quad g = 1 \dots N_G \quad (\gamma_g^G) \quad (1c) \rightarrow \text{Generation limits}$$

$$0 \leq p_d^D \leq \bar{P}_d^D \quad d = 1 \dots N_D \quad (\gamma_d^D) \quad (1d) \rightarrow \text{Consumption limits}$$

- λ and γ are sets of Lagrange multipliers associated with all equality and inequality constraints
- And equivalently as a minimization problem by minimizing the opposite objective function, i.e.

$$\begin{aligned} \text{Min}_{p_g^G \geq 0, p_d^D \geq 0} \quad SW = & \sum_{g=1}^{N_G} C_g p_g^G - \sum_{d=1}^{N_D} U_d p_d^D \quad (2a) \end{aligned}$$

subject to

$$(1b)-(1d) \quad (2b)$$

Grid access, Congestion management

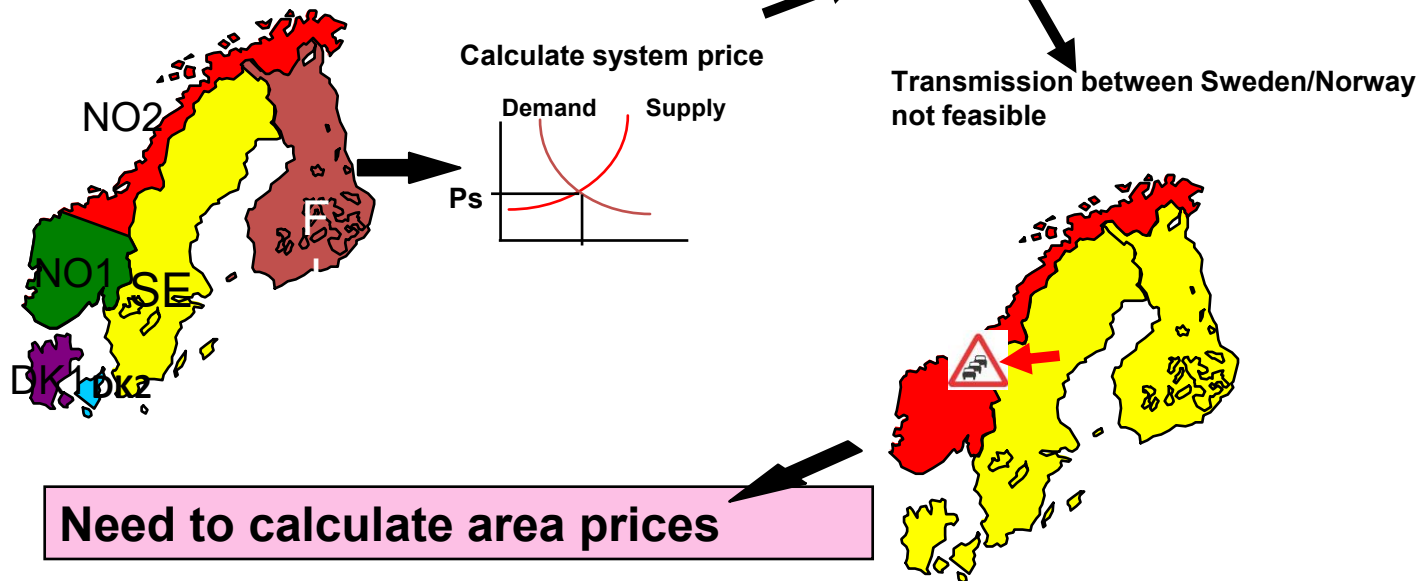
(sources: IW (Ch. 7), Lecture Slides, Assignment 5)

- Understand the principles of congestion management methods, i.e., area pricing and counter trade
- Be able to solve simple cases of area pricing and counter trades in practice
 - Understand calculation of producer and consumer surplus in a single area
 - Understand how these are influenced by interconnection of areas and how this affects the total surplus through the producer and consumer surpluses and the congestion rent
- Understand the economic consequence of congestion (loss of social welfare)
- Understand advantages and disadvantages of each method

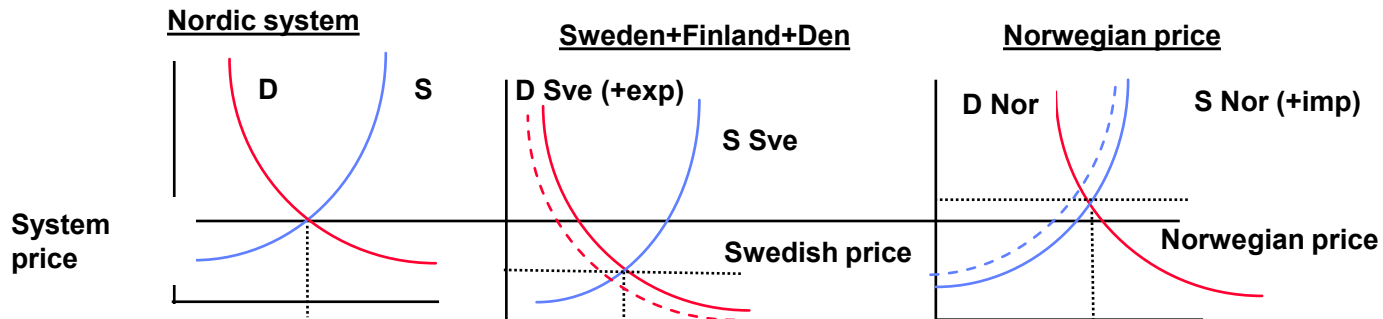
IW: Ivar Wangensteen, “Power System Economics: The Nordic Electricity Market”, Second Edition, 2012.

Area pricing

Example: Nordic system



Source: Nord Pool



System price $\neq$ Area price

Formulating the market clearing

- The grid-constrained social welfare maximization problem can be written as:

$$\text{Max}_{p_g^G, p_d^D, T} \quad SW = \underbrace{\sum_{d=1}^{N_D} U_d p_d^D}_{\text{Utility of demands}} - \underbrace{\sum_{g=1}^{N_G} C_g p_g^G}_{\text{Cost of electricity production}}$$

subject to

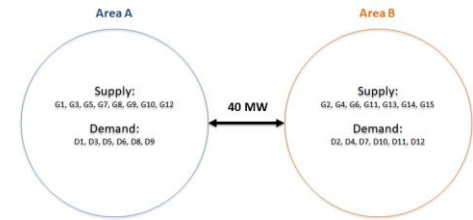
$$\sum_{d=1}^{N_D} p_d^{D,A} - \sum_{g=1}^{N_G} p_g^{G,A} = T \quad \rightarrow \quad \text{Power balance in area A}$$

$$\sum_{d=1}^{N_D} p_d^{D,B} - \sum_{g=1}^{N_G} p_g^{G,B} = -T \quad \rightarrow \quad \text{Power balance in area B}$$

$$0 \leq p_g^G \leq \bar{p}_g^G \quad g = 1 \dots N_G \quad \rightarrow \quad \text{Generation limits}$$

$$0 \leq p_d^D \leq \bar{p}_d^D \quad d = 1 \dots N_D \quad \rightarrow \quad \text{Consumption limits}$$

$$-40 \leq T \leq 40 \quad \rightarrow \quad \text{Transmission limits}$$



where:

- T is the available transfer capacity of the interconnection between area A and area B

Transmission pricing based on OPF

(sources: IW (Ch.7), Lecture Slides, Assignment 6, Course Project, Memo: Use of Lagrange-multipliers, Memo: DC Power Flow in Power Markets, Lecture Notes: “ Chapter 7 OPF”)

- Understand the meaning of the Lagrangian function representing the production cost minimization and the constraints that result from the load flow equations, $g(\mathbf{x}, \mathbf{P}) = 0$, as well as the grid constraints, $h(\mathbf{x}) = 0$, and the constraints on maximum generation
- Be able to solve the problem for small examples both for a DC grid and a DC load flow in an AC grid
- Understand how the power prices are calculated in the OPF context
- Understand the calculation of the loss factors and how they are used to calculate the grid tariffs
- Understand how constraints affect prices in an OPF – nodal prices context

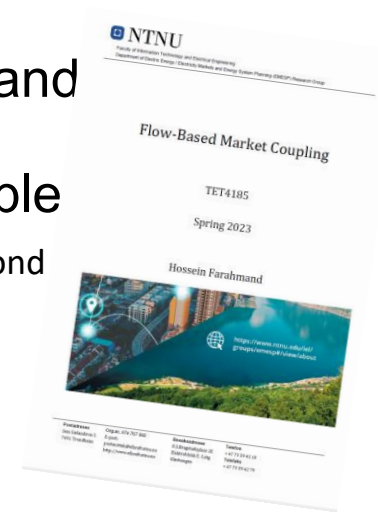
IW: Ivar Wangensteen, “Power System Economics: The Nordic Electricity Market”, Second Edition, 2012.

Grid-constrained market clearing & Flow-Based Market Coupling

(Source: Lecture Slides, Assignment 9, Memo: “Flow-based market coupling”)

- The Grid Is Not a Copper Plate: Markets clear subject to physics.
- From ATC to Flow-Based:
 - ATC → simplified commercial limits
 - DC-OPF → full physical representation
 - PTDF → scalable mapping from injections to flows
 - Flow-Based → physics embedded in zonal markets
- Understand and be able to explain the basics of the Flow-Based method
- Understand the calculation of the PTDFs and be able to use the PTDFs to calculate inter area flows
- Understand the concept of the Flow-Based domain and difference with the ATC domain
- Be able to calculate the Flow-Based domain for simple

IW: Ivar Wangensteen, “Power System Economics: The Nordic Electricity Market”, Second Edition, 2012.



Market clearing considering network

$$\text{Max}_{p_g^G, p_d^D, \theta_n} SW = \sum_{d=1}^{N_D} U_d p_d^D - \sum_{g=1}^{N_G} C_g p_g^G$$

subject to

$$0 \leq p_d^D \leq \bar{P}_d^D \quad d = 1 \dots N_D$$

$$0 \leq p_g^G \leq \bar{P}_g^G \quad g = 1 \dots N_G$$

$$\sum_{d \in \Psi_n} p_d^D - \sum_{g \in \Psi_n} p_g^G + \sum_{m \in \Omega_n} B_{n,m} (\theta_n - \theta_m) = 0 \quad (\lambda^n) \quad \forall n$$

$$-T_{n,m} \leq B_{n,m} (\theta_n - \theta_m) \leq T_{n,m} \quad \forall n, \forall m$$

$$\theta_{ref} = 0$$

All demands
located at bus n

All generators
located at bus n

All buses m connected
to bus n through
transmission lines

Nodal price at bus n
(locational marginal price, LMP)

Market clearing considering network

$$\text{Max}_{p_g^G, p_d^D, \theta_n} SW = \sum_{d=1}^{N_D} U_d p_d^D - \sum_{g=1}^{N_G} C_g p_g^G$$

subject to

$$0 \leq p_d^D \leq \bar{P}_d^D \quad d = 1 \dots N_D$$

$$0 \leq p_g^G \leq \bar{P}_g^G \quad g = 1 \dots N_G$$

$$\sum_{d \in \Psi_n} p_d^D - \sum_{g \in \Psi_n} p_g^G + \sum_{m \in \Omega_n} B_{n,m} (\theta_n - \theta_m) = 0 \quad (\lambda^s) \quad \forall n$$

$$-T_{n,m} \leq B_{n,m} (\theta_n - \theta_m) \leq T_{n,m} \quad \forall n, \forall m$$

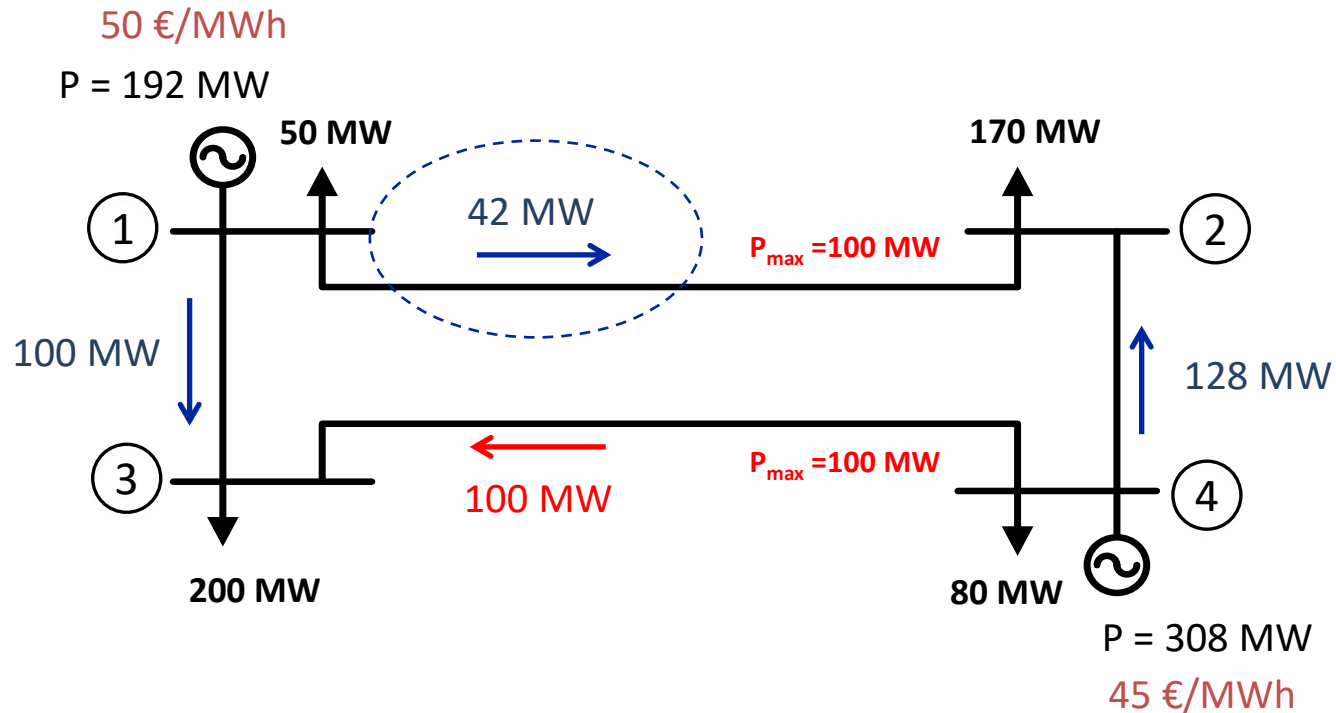
$$\theta_{ref} = 0$$

Voltage angle at the
reference bus

Capacity of line
connecting bus n to bus m

What will we be covering in this 2-part special?

- How does electricity flow in the power grid?
- How can properties of the electricity grid make indirect bottlenecks?



Flow-Based Market Model

Decision variables

- P_g^G, P_d^D : generation and demand dispatch
- NP_z : Net position of bidding zone z

$$\text{Max}_{p_g^G, p_d^D, NP_z} SW = \sum_{d=1}^{N_D} U_d p_d^D - \sum_{g=1}^{N_G} C_g p_g^G \longrightarrow \text{Objective}$$

subject to

$$0 \leq p_d^D \leq \bar{P}_d^D \quad d = 1 \dots N_D \longrightarrow \text{Generation limits}$$

$$0 \leq p_g^G \leq \bar{P}_g^G \quad g = 1 \dots N_G \longrightarrow \text{Demand limits}$$

$$NP_z = \sum_{g=1}^{N_G} p_g^G - \sum_{d=1}^{N_D} p_d^D \quad (\lambda_z) \quad \forall z \in Z \longrightarrow \text{Zonal balance}$$

$$\sum_{z=1}^Z NP_z = 0 \quad (\lambda_s) \longrightarrow \text{Global balance}$$

$$-RAM_k \leq \sum_{z=1}^Z PTDF_{k,z} NP_z \leq RAM_k \quad (\mu_k) \quad \forall k \in K \longrightarrow \text{Transmission limits}$$

$z \in Z$: bidding zones & $k \in K$: critical network elements (CNECs / monitored lines)

Lagrangian of the Flow-Based Problem

Take the simplified welfare maximization:

max SW
subject to:
Zonal balance
Global balance
Transmission limits (PTDF $\times$ NP $\leq$ RAM)

$$\mathcal{L} = SW + \sum_z \lambda_z \left(NP_z - \sum_{i \in z} (g_i - d_i) \right) + \lambda_s \left(0 - \sum_z NP_z \right) + \sum_k \mu_k \left(RAM_k - \sum_z PTDF_{k,z} NP_z \right)$$

First-Order Condition for NP_z

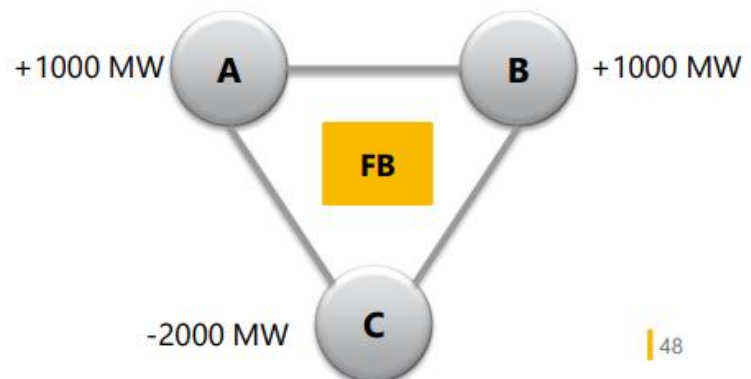
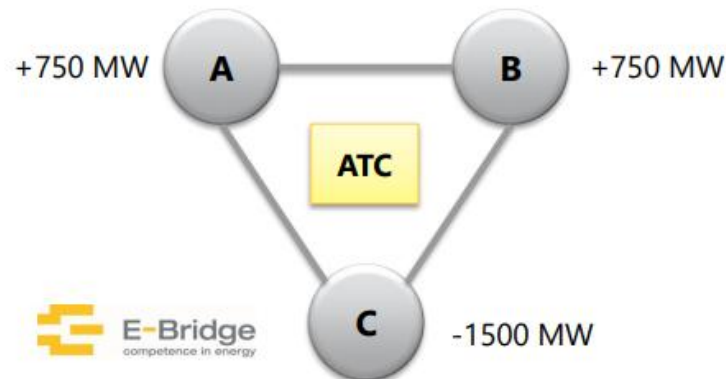
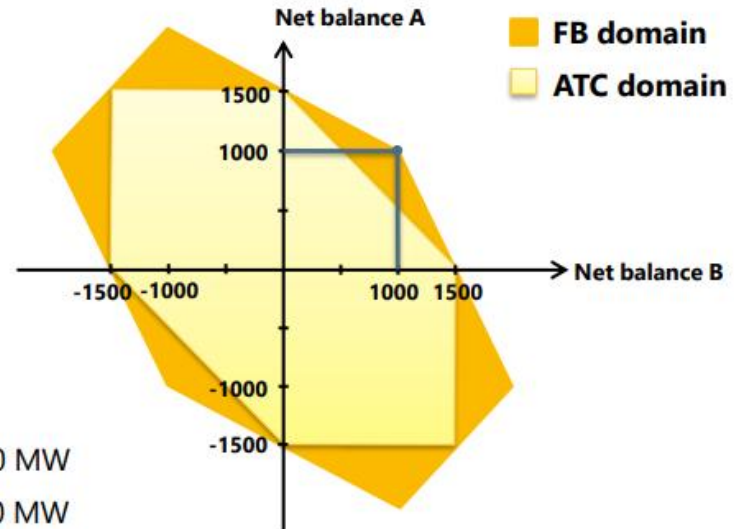
$$\frac{\partial \mathcal{L}}{\partial NP_z} = \lambda_z - \lambda_s - \sum_k \mu_k PTDF_{k,z} = 0 \quad \Rightarrow \quad \lambda_z = \lambda_s + \sum_k \mu_k PTDF_{k,z}$$

Economic Interpretation:

Zonal price = System price + weighted sum of congestion prices
Weights = PTDFs.

An example three-node network: ATC vs FB domain

- In FB capacity split is not a choice of the TSO, but is market driven (at the time of allocation)
- FB offers more trading opportunities with the same level of security of supply
- Example:
 - ATC: North-South exchange limited to 1500 MW
 - FB: North-South exchange possible of 2000 MW



Environmental Policy

(sources: IW (Ch. 10), Lecture Slides, Assignment 7)

Learning objectives:

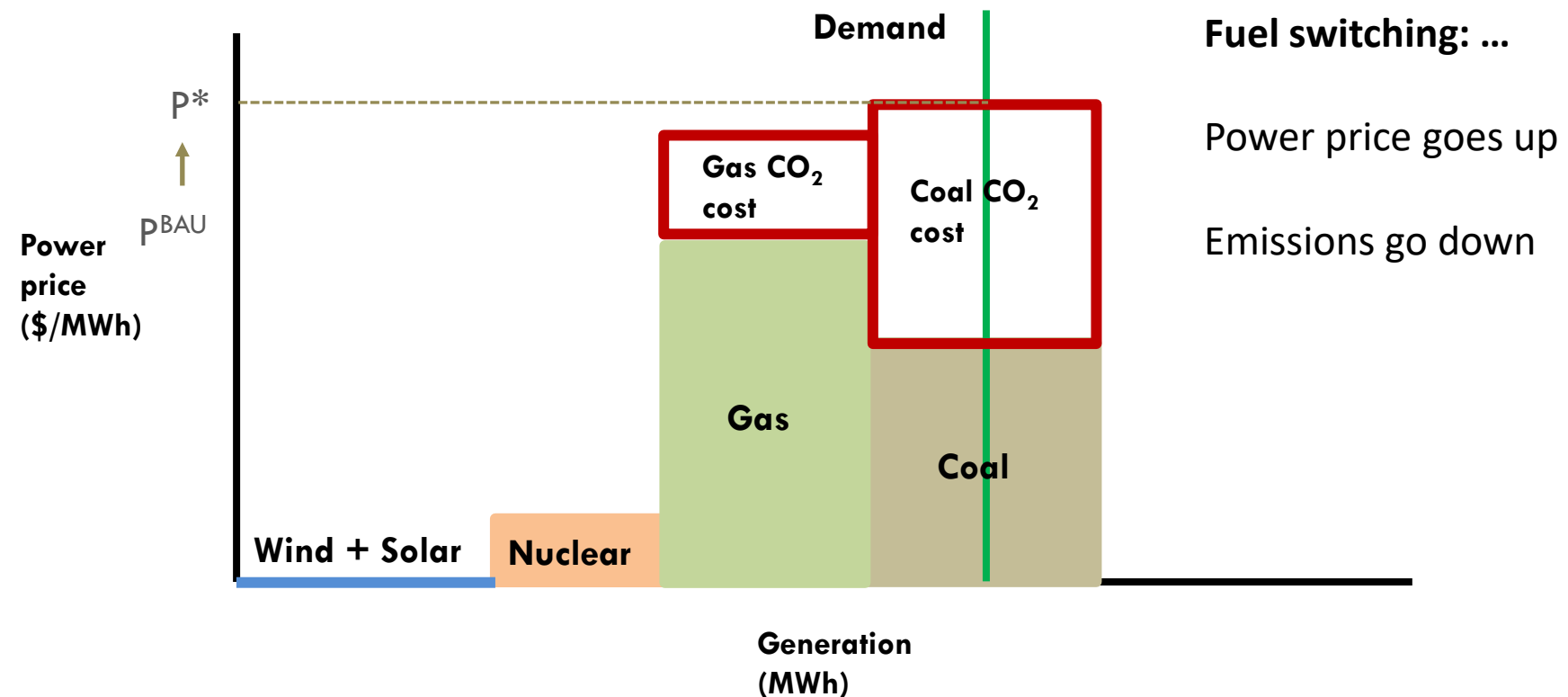
- Understanding fundamental concepts and frameworks in climate policy
- Familiarity with tools for modeling climate policy

The Contents

- Theoretical foundations of environmental policy
- Climate policy and CO2 pricing
- Cap-and-trade, a.k.a. Emissions Trading Systems
- The EU Emissions Trading System
- Technology policy (renewable energy policy)
 - Green certificate schemes
 - feed-in-tariff (FIT)
 - fixed feed-in premium (FIP)
 - contract for difference (CfD, or sliding premium)

IW: Ivar Wangensteen, “Power System Economics: The Nordic Electricity Market”, Second Edition, 2012.

The new market equilibrium:



Modeling cap-and-trade

The commonly used way to model cap-and-trade (emissions trading) in optimization:

$$\max_{p_g^G, p_d^D} \sum_d^{N_D} U_d p_d^D - \sum_g^{N_G} C_g p_g^G$$

$$\text{s.t.} \quad \sum_d^{N_D} p_d^D - \sum_g^{N_G} p_g^G = 0 \quad (\lambda) \quad \leftarrow \text{Power balance}$$

$$p_g^G \leq \bar{P}_g^G \quad (\mu_g) \quad \forall \quad g \quad \leftarrow \text{Capacity limit}$$

$$p_d^D \leq \bar{P}_d^D \quad (\gamma_d) \quad \forall \quad d$$

$$\text{Cap on CO}_2 \rightarrow \sum_g E_g^{CO_2} p_g^G \leq \bar{E}^{CO_2} \quad (\kappa) \quad \uparrow \text{CO}_2 \text{ price}$$

Variables:

p_g^G : generation of technology g (MWh)

p_d^D : demand of bid g (MWh)

Parameters:

C_g^G : Variable cost of technology g (\$/MWh)

U_d^D : Utility (i.e. maximum willingness to pay) for bid d (\$/MWh)

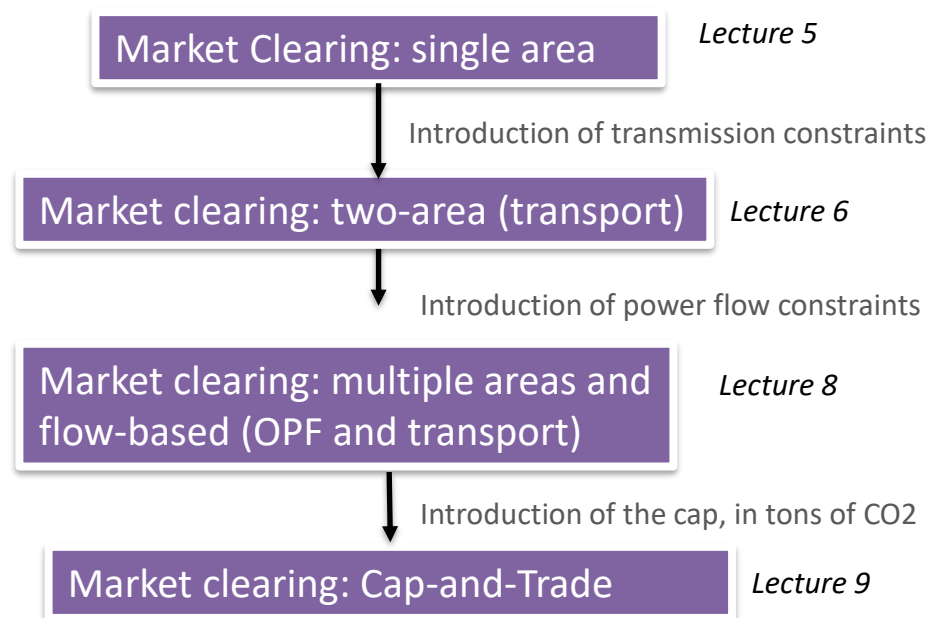
$\bar{P}_g^G$: Capacity of technology g (MW)

$E_g^{CO_2}$: Emissions intensity of technology g (tCO₂/MWh)

$\bar{E}^{CO_2}$: Cap on emissions (tCO₂)

Modelling market clearing as an optimization problem

- **Objective:** Model and solve market clearing in power systems using optimization techniques.
- **Approach:** Emphasis on optimization coding to develop practical skills in formulating and solving complex market models.
- Students will be able to model various market clearing scenarios as optimization problems, implement these models in code, and critically analyze the outcomes to make informed decisions in the context of power economics.



Hydro Generation

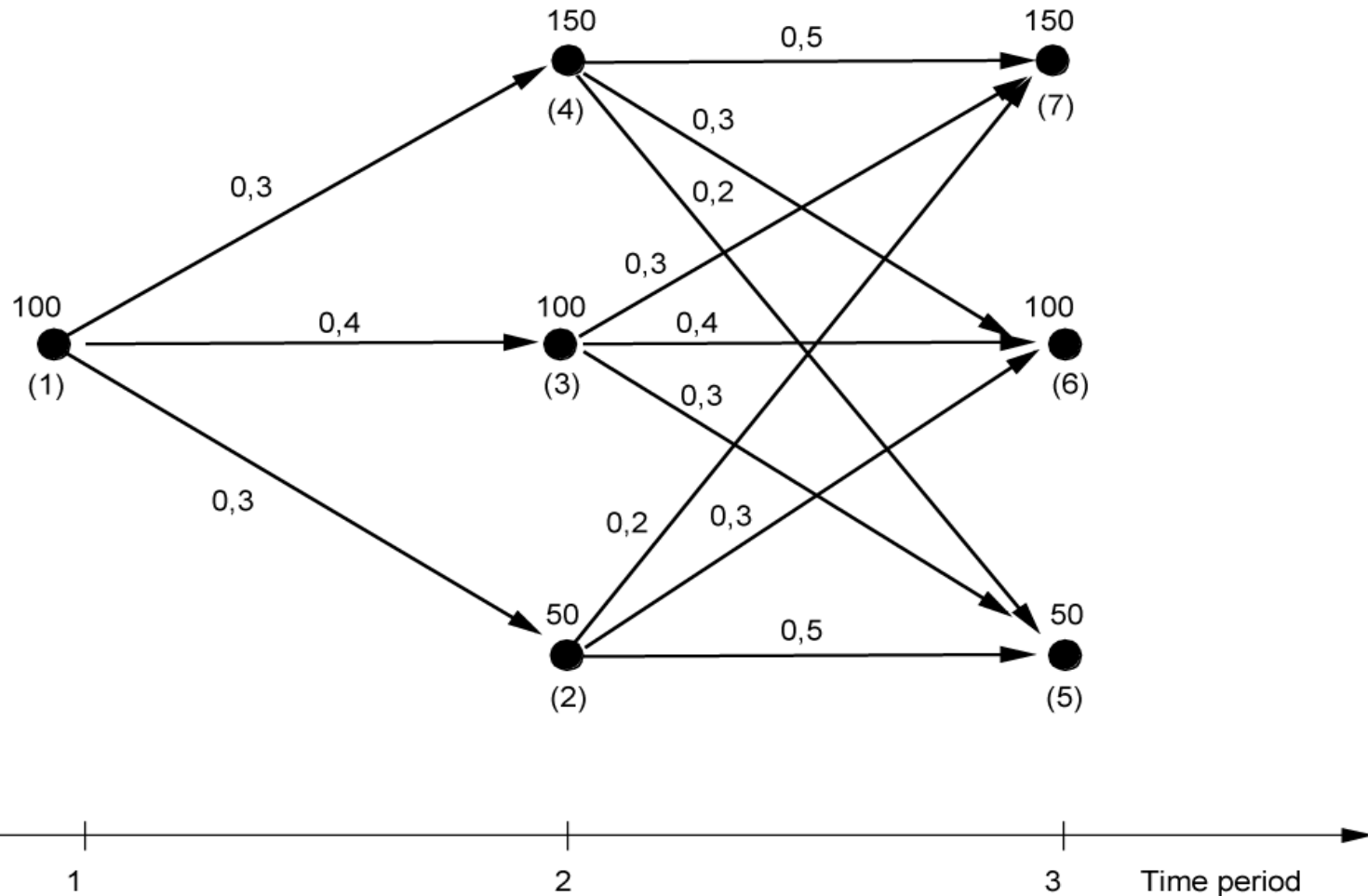
(sources: IW (Ch. 6), Lecture Slides, Assignment 8)

- Understand the basic elements of hydro scheduling
- Understand the bathtub model and be able to do simple calculations based on the model
- Be able to use a simple price model for hydro scheduling based on scenario tree for possible future market price

IW: Ivar Wangensteen, “Power System Economics: The Nordic Electricity Market”, Second Edition, 2012.

Example: “Recombining” Scenario tree for possible future market price

We have hydro plant with reservoir capacity of 100 GWh



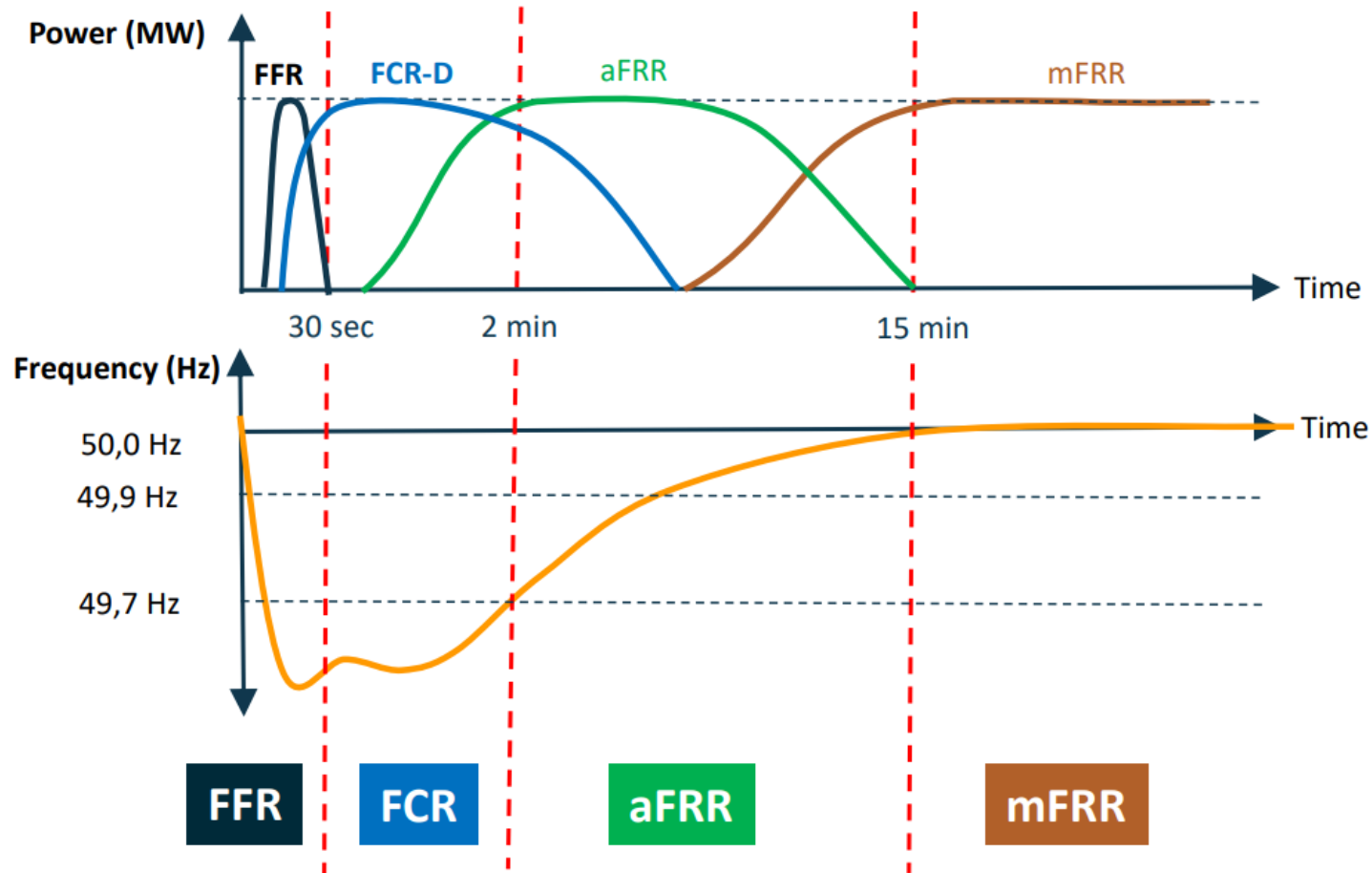
Ancillary services

(sources: IW (Ch. 8), Lecture Slides)

- Define ancillary services and explain their necessity for maintaining power system integrity and stability
- Explore frequency-supporting services, including Frequency Containment Reserve (FCR), Frequency Restoration Reserve (FRR), and Replacement Reserves (RR).
- Understand and be able to use unit droop in simple calculations
- Understand the main operational reserve parameters
- Analyze the mechanisms and market operations for ancillary services within the Nordic system and upcoming pan-European platforms

IW: Ivar Wangensteen, “Power System Economics: The Nordic Electricity Market”, Second Edition, 2012.

Frequency supporting Ancillary services

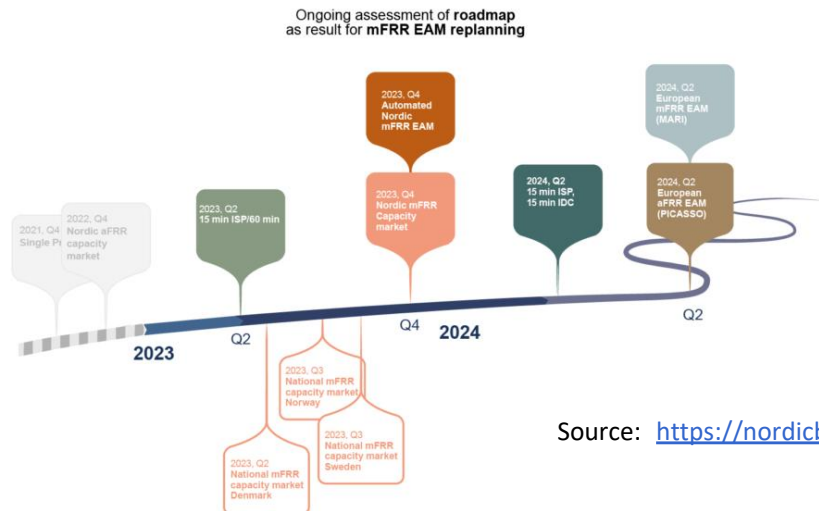


Upcoming pan-European balancing platforms

- ▶ **PICASSO**: An exchange platform for **aFRR** between European countries (https://www.entsoe.eu/network_codes/eb/picasso/)
- ▶ **MARI**: An exchange platform for **mFRR** between European countries (https://www.entsoe.eu/network_codes/eb/mari/)

The expected go-live date for these European balancing platforms is in Q2, 2024.

Statnett has a close operation and market cooperation with the other Nordic TSO's and they have decided to implement a common project called *Nordic Balancing Model*, **NBM** (<https://nordicbalancingmodel.net/>) to eventually join PICASSO and MARI.



Source: <https://nordicbalancingmodel.net/roadmap-and-projects/>

Final comment

- Please read Learning Objectives and Study Guide TET4185
- You can find the document under “Course information” on Blackboard

